SA6210A XA5 Signal Analyzer

2 Hz to 32 GHz

Introduction

This SA6210A configuration guide will help you determine which performance options, measurement applications, accessories, and services to include with your new SA6210A order or to add as upgrades to an existing SA6210A.



KEYSIGHT

Included in Base Product

The options and accessories are included as standard on the SA6210A XA5 Signal Analyzer base instrument at no additional charge and do not need to be ordered separately.

They include:

- KeysightCare Warranty Plus First Year Support (includes Return to Keysight Warranty – 1 year)
- Quick start poster
- Spectrum analyzer software application
- Low phase noise
- 25 MHz analysis bandwidth
- Microwave preselector bypass
- Noise floor extension
- i7 Processor, 64GB DDR4
- Microsoft Windows 11 operating system
- Multi-language user interface
- Multi-touch user interface
- Country-specific power cord

Configure Your SA6210A XA5 Signal Analyzer

This step-by-step process will help you to configure the SA6210A XA5 signal analyzer. Tailor the performance and features to meet your requirements. For detailed specifications, please refer to the SA6210A XA5 Signal Analyzer Data Sheet (3126-1370.EN).

Step 1. Select Number of Channels (Required Option)

Description	Option Number	Additional Information
Add channel one	SA6210A-001	Required for all configurations
Add channel two	SA6210A-002	

Step 2. Select Maximum Frequency Range

Description	Option Number	Additional Information
Frequency range, 2 Hz to 32 GHz, RF 1	SA6210A-532	For channel one
Frequency range, 2 Hz to 32 GHz, RF 2	SA6210A-632	For channel two

Step 3. Add a Preamplifier

Description	Option Number	Additional Information
Preamplifier 32 GHz, RF 1	SA6210A-P32	Compatible with SA6210A-532 only; for channel one
Preamplifier 32 GHz, RF 2	SA6210A-Q32	Compatible with SA6210A-632 only; for channel two

Step 4. Select Analysis Bandwidth

Description	Option Number	Additional Information
Analysis bandwidth, 25 MHz, RF1	Standard	Licensed as SA6210A-B25; for channel one
Analysis bandwidth, 25 MHz, RF2	Standard	Licensed as SA6210A-C25; for channel two
Analysis bandwidth, 40 MHz, RF 1	SA6210A-B40	For channel one
Analysis bandwidth, 40 MHz, RF 2	SA6210A-C40	For channel two
Analysis bandwidth, 255 MHz, RF 1	SA6210A-B2X	For channel one
Analysis bandwidth, 255 MHz, RF 2	SA6210A-C2X	For channel two
Analysis bandwidth, 480 MHz, RF 1	SA6210A-B4X	For channel one
Analysis bandwidth, 480 MHz, RF 2	SA6210A-C4X	For channel two
Analysis bandwidth, 1 GHz, RF 1	SA6210A-R10	For channel one
Analysis bandwidth, 1 GHz, RF 2	SA6210A-S10	For channel two
Analysis bandwidth, 2 GHz, RF 1	SA6210A-R20	For channel one
Analysis bandwidth, 2 GHz, RF 2	SA6210A-S20	For channel two

Step 5. Add a Preselector

Description	Option Number	Additional Information
Add tunable preselector 32 GHz, RF 1	SA6210A-YF3	Compatible with SA6210A-532 only; for channel one
Add tunable preselector 32 GHz, RF 2	SA6210A-SY3	Compatible with SA6210A-632 only; for channel two

Step 6. Add Performance Options

Description	Option Number	Additional Information
Low phase noise performance	Standard	Licensed as SA6210A-EPB
Microwave preselector bypass	Standard	Bypass the microwave preselector for wider bandwidth IF; licensed as SA6210A-MPB
Noise floor extension	Standard	Improves analyzer's DANL performance; licensed as SA6210A-NF2
Enhanced display package	N90EMEDPB	Includes spectrogram, trace zoom, and zone span in SA mode
Resolution Bandwidth extended, basic	N90EMWR1B	Extends the maximum RBW to 40 MHz; requires option B2X, B4X, R10, or R20; non-IVL, no export control
Resolution bandwidth extended, optimum	N90EMWR2B	Extends the maximum RBW to 80 MHz; requires Option B2X, B4X, R10, or R20
Multi-channel measurement	SA6210MCHB	Enables two-channel measurements (e.g. ccEVM, MIMO) in VMA, WLAN, and 5G NR measurement applications; requires Option 001 and 002

Step 7. Add Security Features

Description	Option Number	Additional Information
Additional removable Solid-State Drive (SSD)	SA6210A-SD1	Provides a fully imaged, removable SSD in addition to the one installed in instruments, with Windows 11 operating system
Security features, exclude launching programs	SA6210A-SF1	Prevents the launching of Windows programs from the instrument application
Security features, prohibit saving results	SA6210A-SF2	Prevents instrument application from saving/recall of measurement results or user configurations to/from instrument's data storage
Security feature, secure RAM disk boot	SA6210A-SF3	Allows the instrument to boot the Windows OS into volatile memory, which ensures that, after a power cycle, all information from the previous boot is lost

Step 8. Add Connection Options

Description	Option Number	Additional Information
Connector, GPIB, Micro-D 25-pin	SA6210A-GPB	
Connector, 10 Gbits Ethernet port	SA6210A-GBE	

Step 9. Select Measurement Applications and License Type

Note: Keysight offers flexible license types and terms for the measurement applications, refer to X-Series Measurement Applications – Brochure (5989-8019EN).

Description	Option Number	Additional Information
General Purpose		
Spectrum analyzer and IQ analyzer	Standard	Traditional spectrum analysis plus many new and enhanced functions; licensed as N9060ES1E
Power Suite	N90EMPSMB	Power measurements based on industry specifications
Vector modulation analysis digital demodulation	N9054EM0E	Performs one-button flexible modulation analysis measurements with FSK, PSK, QAM, MSK, ASK, APSK, VSB etc. and popular format preset
Vector modulation analysis custom OFDM	N9054EM1E	Performs one-button custom OFDM modulation analysis measurement with user-defined settings or recalling 89600 VSA or Signal Studio output files

Phase noise	N9068EM0E	Adds one-button measurements for analyzing phase noise in frequency domain (log plot) and time domain (spot frequency), supports external mixing
Wireless Connectivity		
WLAN 802.11a/b/g/j/p/n/af/ah	N9077EM0E	Standard-based, one-button 802.11a/b/g/j/p/n/af/ah measurement
WLAN 802.11ac/ax	N9077EM1E	Standard-based, one-button 802.11ac/ax measurement
WLAN 802.11be/bn	N9077EM2E	Standard-based, one-button WLAN 802.11be/bn measurement
Cellular Communications		
5G NR/5G-Advanced	N9085EM0E	Standard-based, one-button 5G NR (New Radio) and 5G-Advanced downlink and uplink measurements
NR V2X	N9085EM4E	Standard-based, one-button NR V2X transmitter measurements

Step 10. Select 89600 VSA Software Licenses

Description	Option Number	Additional Information
Advanced vector signal analysis	89601201C	Advanced vector signal analysis all-inclusive with no frequency, bandwidth limits and up to 64-channel and 512 parallel instance support
Standard vector signal analysis	89601202C	Standard vector signal analysis up to 55 GHz center frequency, 2.16 GHz bandwidth, and 4 channels and 4 parallel instances support
Essential vector signal analysis	89601203C	Essentials vector signal analysis up to 8 GHz center frequency, 160 MHz bandwidth, and single channel support
General Purpose		
Power Suite	89601PSMC	PowerSuite measurement for ACP and SEM
Digital demodulation analysis	89601AYAC	Analysis of > 40 modulation formats for communication formats like GSM/EDGE/W-CDMA, ZigBee, Wi-SUN, Bluetooth BR, WLAN, APCO25, J.83A/B/C, DVB-S2, ATSC, TETRA, DECT, and TEDS
		Perform modulation analysis on non-standard user-defined IQ signals through Custom IQ measurement Perform modulation analysis with flexible frame structure definition of multiple segments of preamble, pilot, data, and idle through FlexFrame measurement
Custom OFDM modulation analysis	89601BHFC	Proprietary and pre-standard OFDM formats modulation analysis such as 6GR (FR1/FR3), 5G NR (FR1), WLAN, DAB, DVB-T/H, DVB-SH, ISDB-T and more

Direct data connectivity	89601101C	Push live IQ data through API programming (.NET or SCPI) or VITA-49 streaming over LAN without the use of supported hardware or recording files
ccEVM and IQ-NC	89601EVMC	Improve EVM measurement through ccEVM or IQ-NC technologies
Cellular Communications		
3G modulation analysis	89601B7NC	W-CDMA/HSPA+, TD-SCDMA/HSPA, cdma2000, 1xEV-DO and 1xEV-DV modulation analysis
LTE/LTE-A FDD modulation analysis	89601BHGC	LTE/LTE-Advanced FDD modulation analysis
LTE/LTE-A TDD modulation analysis	89601BHHC	LTE/LTE-Advanced FDD modulation analysis
5G NR/5G-Advanced modulation analysis	89601BHNC	5G NR/5G-Advanced, Pre-5G modulation analysis
Stimulus response signal analysis	89601CSDC	Performs channel sounding measurement Perform advanced component test features inside component test measurement
O-RAN ULPI	89601ULPC	Supports ULPI (Uplink Performance Improvement) for O-RAN to enhance uplink massive MIMO
Phase Coherent Channel Count Expander Up to 8-Port	89601CC1C	Phase coherent channel count expander up to 8-Port with Sequential Acquisition
Phase Coherent Channel Count Expander Up to 64-Port	89601CC2C	Phase coherent channel count expander up to 64-Port with Sequential Acquisition
Wireless Connectivity		
Wireless connectivity modulation analysis	89601B7RC	WLAN 802.11a/b/g/j/p modulation analysis WiMax modulation analysis (including both EVM and SEM)
High throughput WLAN modulation analysis	89601BHXC	WLAN 802.11n/ac/ax/be/bn modulation analysis (including both EVM and SEM)
IoT modulation analysis	89601BHTC	NB-IoT, RFID, HRP UWB (802.15.4/4z/4ab) modulation analysis
Radar Analysis		
FMCW radar analysis	89601BHPC	Analyze Frequency Modulated Continuous Wave (FMCW) radar signals used in automotive radar, industrial, surveillance, aerospace and defense radar applications
Pulse analysis	89601BHQC	Gain deeper insights to FM linear chirp, frequency hopping, and pulsed radar signals
Other Standard Formats		
DOCSIS modulation analysis	89601BHMC	DOCSIS3.1/4.0 downstream and upstream modulation analysis
Satellite communication analysis	89601DVBC	Full channel decoding and modulation analysis for DVB-S2/DVB-S2X (including supper Frame 5/6/7) and DVB-RCS2 return link modulation analysis

Step 11. Select Accessories

Description	Option Number	Additional Information
Quick start poster	Standard	US – English localization
Power cord	Standard	Dependent upon the region of use
Adapter, 2.4 mm (f) to 2.92 mm (f)	Standard	For Option 532 and 632
Front and rear panel covers	CV1118A	Protective covers for front and real panels
USB DVD-ROM/CD-R/RW drive	1DVR001A	Enhances the usability of the Windows OS
USB mouse	1MSE001A	Enhances the usability of the 89600 VSA software
GPIB cable	Y1260A	Connects a MicroD25 GPIB to a standard GPIB
Rack mount kit, without handles	SA6210A-2CM	Rack mount flange kit – two flange brackets, two standard rails
Rack mount kit, with handles	SA6210A-2CP	Rack mount flange and handle kit – two flange brackets, handles, rails

For more information on accessories, please go to: www.keysight.com/find/accessories

Step 12. Order Calibration, Service and Support

Description	Option Number	Additional Information
KeysightCare Warranty Plus first year support	R-55F-001-1	Includes Return to Keysight Extended Warranty
KeysightCare Upgrades – Warranty and Calibration		
KeysightCare Warranty Plus – Extend to 2 years	R-55F-001-2-D	Includes Return to Keysight Extended Warranty
KeysightCare Warranty Plus – Extend to 3 years	R-55F-001-3-D	
KeysightCare Warranty Plus – Extend to 5 years	R-55F-001-5-D	
Calibration Services		
Commercial calibration certificate with test data	SA6210A-UK6	Calibration certificate only available at time of instrument purchase; only provides measurement results
Calibration Plan – Return to Keysight – 3 years	R-50C-011-3-D	
Calibration Plan – Return to Keysight – 5 years	R-50C-011-5-D	
Product Assistance		
Remote scheduled productivity assistance	PS-S10	
Startup assistance, daily	PS-S20	
Custom services	PS-X10	

Other calibration options may be available; for more information on calibration go to: www.keysight.com/find/calibration

Instrument Upgrades

Fast license-key upgrades for options that do not require additional hardware:

1. Place an order for the upgrade with Keysight and request to receive the option upgrade entitlement certificate and a one-time software upgrade license through email
2. Redeem the certificate through the Web by following the instructions on the certificate
3. Install the license file and latest software in the SA6210A
4. Begin using the new capability [1, 2]

Description	Upgrade Number	Requirements	Additional Information
Analysis Bandwidth			
Increase channel 1 analysis bandwidth from 25 MHz to 40 MHz (B40)	SA6210AU-B01	B25	License key only; It is required that analysis bandwidths for channel 1 and channel 2 are identical
Increase channel 2 analysis bandwidth from 25 MHz to 40 MHz (C40)	SA6210AU-C01	C25	License key only; It is required that analysis bandwidths for channel 1 and channel 2 are identical
Increase channel 1 analysis bandwidth from 255 MHz to 480 MHz (B4X)	SA6210AU-B10	B2X	License key only; It is required that analysis bandwidths for channel 1 and channel 2 are identical
Increase channel 2 analysis bandwidth from 255 MHz to 480 MHz (C4X)	SA6210AU-C10	C2X	License key only; It is required that analysis bandwidths for channel 1 and channel 2 are identical
Increase channel 1 analysis bandwidth from 1 GHz to 2 GHz (R20)	SA6210AU-B15	R10	License key only; It is required that analysis bandwidths for channel 1 and channel 2 are identical
Increase channel 2 analysis bandwidth from 1 GHz to 2 GHz (S20)	SA6210AU-C15	S10	License key only; It is required that analysis bandwidths for channel 1 and channel 2 are identical
Additional Upgrades			
Add security features, exclude launching programs	SA6210AU-SF1	None	License key only
Add security features, prohibit saving results	SA6210AU-SF2	None	License key only
Add security features, secure RAM disk boot	SA6210AU-SF3	None	License key only
Add additional removable solid state drive, Win11	SA9094BU-SD1 (under SA6210AU)	None	Provides additional removable solid-state drive, with Windows 11 operating system

1. At the time of manufacture, the hardware related to many of these options was fully adjusted and the option performance was verified to be within its warranted specifications. Within one year of the initial calibration date of the analyzer, this option is fully calibrated with no further adjustment or verification testing.
2. If this analyzer has been adjusted as part of a repair or calibration during its first year, or if the analyzer is more than one year old, additional adjustment and performance verification tests are required to ensure that some newly installed options are functioning properly. However, the completion of these tests does not guarantee that the analyzer meets all warranted specifications.

Keysight Support Services

Accelerate your learning curve, enhance your test uptime, and confidently guarantee your instrument accuracy with Keysight Support Services. Keysight Services are here to support your test needs with expert technical support, instrument repair and calibration, training, alternative acquisition program options, and more.

A KeysightCare agreement provides dedicated, proactive support through a single point of contact for an extensive group of instruments, software, and solutions to ensure optimal uptime, with fast response times and resolution. Explore the services that are right for you.

Keysight Services

Offering	Benefits
KeysightCare KEYSIGHTCARE	KeysightCare provides elevated support for Keysight instruments and software, with access to technical support experts who respond within a specified time and ensure committed repair and calibration turnaround times (TAT). KeysightCare offers multiple service agreement tiers, including KeysightCare Assured, Enhanced, and Application Software Support. See the KeysightCare data sheet for details.
KeysightCare Assured	KeysightCare Assured provides a commitment to respond to your engineer's technical needs quickly. When unexpected repairs are necessary, you can count on a committed repair service turnaround time to get you back up and running.
KeysightCare Enhanced	KeysightCare Enhanced includes all the benefits of KeysightCare Assured plus Keysight's accurate and reliable Calibration Services, accelerated and committed TAT, and technical response.
Keysight Support Portal & Knowledge Center	All KeysightCare tiers include access to the Keysight Support Portal, where you can manage support and service resources related to your assets, such as service requests and status, or browse the Knowledge Center.
Education Services	Build confidence and gain new skills to make accurate measurements, with flexible Education Services developed by Keysight experts. Including Start-up Assistance.
Alternative Acquisition Options	
KeysightAccess	Reduce budget challenges with a lease-based subscription service that offers low monthly payments, enabling you to get the instruments, software, and technical support you want for your test needs.

Recommended Services

Maximize your instrument uptime and confidently make accurate measurements by securing technical support, repair, and calibration services with committed response and turnaround times. High-performance instruments include 1 year of KeysightCare Assured or KeysightCare Warranty Plus. Obtain multi-year KeysightCare upfront to eliminate the need for lengthy and tedious paperwork and yearly requests for maintenance budget. Plus, you benefit from secured service for 2, 3, or 5 years.

Service	Function
KeysightCare Enhanced*	**Includes Tech Support, Warranty and Calibration**
R-55B-001-1	KeysightCare Enhanced – Upgrade 1 year
R-55B-001-2	KeysightCare Enhanced – Extend to 2 years
R-55B-001-3	KeysightCare Enhanced – Extend to 3 years (Recommended)
R-55B-001-5	KeysightCare Enhanced – Extend to 5 years (Recommended)
KeysightCare Assured*	**Includes Tech Support and Warranty**
R-55A-001-2	KeysightCare Assured – Extend to 2 years
R-55A-001-3	KeysightCare Assured – Extend to 3 years
R-55A-001-5	KeysightCare Assured – Extend to 5 years
Start-Up Assistance	
PS-S40-01	Included – instrument fundamentals and operations starter
PS-S40-04	Recommended – instrument fundamentals and operations starter
PS-S40-02	Optional, technology & measurement science standard learning

* Limited availability might apply. Please review the service definition tool for model number availability and the datasheet for country availability. Coverage might be limited to KeysightCare Warranty Plus (R-55F-001). If KeysightCare Enhanced is available. R-55B-001-2/3/5 must be ordered with R-55B-001-1.

Keysight enables innovators to push the boundaries of engineering by quickly solving design, emulation, and test challenges to create the best product experiences. Start your innovation journey at www.keysight.com.

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